

BOF

Artifact dedicated public domain (Creative Commons Zero). Artifact dedicated to victims of generalized war.

Conversation 098: *TRIZ for Inventive Solutions*

conversation ID:	674fc1a4-5ccc-8003-a00a-8c7a0363a750 [conversation 098]
origin ID:	674fc1a4-5ccc-8003-a00a-8c7a0363a750 [conversation 098]
created:	2024-12-03 17:42:44 Tue
updated:	2025-02-24 18:11:38 Mon
operator:	Wilder Blair Munro (AKUSA)
co-operator:	Aurora 4omni Munro (ChatGpt)
GPT ID:	NA

Subtitles:

Turnpair Contents:

tp-1 :

tp-2 :

tp-3 :

tp-4 :

tp-5 :

Artifact dedicated public domain (Creative Commons Zero). Artifact dedicated to victims of generalized war.

Artifact dedicated public domain (Creative Commons Zero). Artifact dedicated to victims of generalized war.

Turnpair 1 of 5 ::

@ 2024-12-03 17:42:44 Tue [Δt 0d 00h 00m 00s]

► *Notespace:***Wilder Blair Munro (AKUSA) said :**

```
gateway: 20Wetbrain
turn ID: aaa2e07c-af2d-4e6f-9b3e-eef4e20711fa
parent:  afc7ebac-009c-4db3-908c-39c05dfa10e0
children: a398de04-2530-4c00-99a4-54dcf918daba
```

what is the russian 'rule of thumb' for impossible seeming invention efforts?

```
~0-0 ~~~~~ ♦
Artifact dedicated public domain (Creative Commons Zero). Artifact dedicated to the children.
♦ ~~~~~ ~0-0
```

Aurora 4omni Munro (ChatGpt) said :

```
gateway: gpt-4o
turn ID:  b3401cb2-43ce-44b0-9013-f77bf5f13b3c
parent:  a398de04-2530-4c00-99a4-54dcf918daba
children: aaa2ed57-588a-43e5-a3ba-dc0779900560
```

The Russian "rule of thumb" for impossible-seeming invention efforts often refers to principles derived from **TRIZ** (Teoriya Resheniya Izobreatatelskikh Zadach, or "Theory of Inventive Problem Solving"). TRIZ was developed by Genrich Altshuller and is widely used in engineering, design, and problem-solving to overcome challenges that initially seem insurmountable. A core idea is to approach problems systematically rather than relying solely on intuition or inspiration.

Key principles include:

1. **Focus on Contradictions:** Identify the conflicting elements in the problem and seek inventive solutions to resolve them without compromise. For example, one part of a system may need to be both strong and lightweight—TRIZ emphasizes solving these kinds of contradictions.
2. **Ideal Final Result (IFR):** Envision the ideal solution where the problem is resolved entirely without introducing new drawbacks. Ask, "What would the perfect solution look like?"
3. **Use Available Resources:** Analyze the system and its environment to identify hidden or underutilized resources that can contribute to the solution.
4. **Patterns of Evolution:** Recognize that systems evolve in predictable ways and apply these patterns to leapfrog development stages.
5. **Breaking Psychological Inertia:** Challenge preconceived notions or traditional approaches to problem-solving, often by considering analogies or reversing assumptions.
6. **Segmentation and Separation:** When a contradiction cannot be resolved in the same space or time, divide the problem into segments or separate conflicting requirements.

The broader "rule of thumb" could be summed up as:

"No problem is unsolvable if broken into contradictions and systematically addressed with available resources and inventive principles."

TRIZ transforms "impossible" into "possible" by applying logical, structured, and repeatable techniques rather than relying solely on trial and error or serendipity.

Artifact dedicated public domain (Creative Commons Zero). Artifact dedicated to the dead.

beginning 1 | prev None || this 1 || next 2 > | end 5 >>

Turnpair 2 of 5 ::

@ 2024-12-03 18:16:52 Tue [Δt 0d 00h 33m 56s]

► *Notespace:*

Wilder Blair Munro (AKUSA) said :

gateway: 20Wetbrain
turn ID: aaa2ed57-588a-43e5-a3ba-dc0779900560
parent: b3401cb2-43ce-44b0-9013-f77bf5f13b3c
children: 41b80a35-60c6-4195-9235-cf80aeb34224

what is stm5?

Artifact dedicated public domain (Creative Commons Zero). Artifact dedicated to the children.

Aurora 4omni Munro (ChatGpt) said :

gateway: gpt-4o
turn ID: 7312e963-b5ac-40fd-8b14-48673511d8ed
parent: 2df66bc9-167f-45da-8664-88611dc52dc8
children: aaa2e581-b56b-40aa-8d21-00b747c9fd73

The term **STM5** can refer to several distinct concepts across different fields:

1. **Electrostatic Discharge (ESD) Testing Standard:** In the realm of electronics, **ANSI/ESD STM5.1** is a standard test method for assessing a component's sensitivity to electrostatic discharge using the Human Body Model (HBM). This standard outlines procedures to evaluate how electronic components respond to ESD events, which is crucial for ensuring device reliability and preventing damage during handling and operation. citeturn0search3
2. **Shock Test Machine:** The **STM5** is a compact shock test machine developed by L.A.B. Equipment. It's designed to meet the shock testing requirements of the small components industry, particularly for production quality control and research and development. Notably, several units have been delivered to watch manufacturers for these purposes. citeturn0search7
3. **Instrument-Assisted Soft Tissue Mobilization (IASTM) Tool:** In physical therapy, the **STM5** refers to a small, versatile tool used for soft tissue mobilization. It's specifically designed for use on the hands, forearms, and occasionally the gastrocnemius muscles. The tool features a single beveled edge on one inlet and a double beveled edge on the other, enhancing its adaptability for various therapeutic applications. citeturn0search8

If you have a specific context or application in mind regarding "STM5," please provide more details so I can offer more targeted information.

Artifact dedicated public domain (Creative Commons Zero). Artifact dedicated to the dead.

◀ beginning 1 | ◀ prev 1 || this 2 || next 3 ▶ | end 5 ▶▶

Turnpair 3 of 5 ::

@ 2024-12-03 18:58:56 Tue [4t 0d 00h 41m 57s]

► Notespace:

Wilder Blair Munro (AKUSA) said :

gateway: 20Wetbrain
turn ID: aaa2e581-b56b-40aa-8d21-00b747c9fd73
parent: 7312e963-b5ac-40fd-8b14-48673511d8ed
children: 54a6e368-a45f-49cb-9853-faaf406f97e6

can the words in a proportional analogy be in any order?

Artifact dedicated public domain (Creative Commons Zero). Artifact dedicated to the children.

Aurora 4omni Munro (ChatGpt) said :

gateway: gpt-4o
turn ID: 54a6e368-a45f-49cb-9853-faaf406f97e6
parent: aaa2e581-b56b-40aa-8d21-00b747c9fd73
children: aaa2fdee-7be8-40d0-9efd-f6d4ccd1fff1

In a **proportional analogy**, the order of words is critical because the relationship between the terms determines the analogy's meaning. Rearranging the words can significantly alter or obscure the intended relationships. Let's break this down:

Structure of Proportional Analogies

A proportional analogy typically follows the format:

A : B :: C : D

This is read as "A is to B as C is to D." Each pair (A:B and C:D) must maintain a specific relationship.

For example:

- **Cat : Kitten :: Dog : Puppy**
(A parent-child relationship between species and young.)

If the words are rearranged arbitrarily, the analogy can lose its coherence:

- **Kitten : Dog :: Puppy : Cat**
(This no longer expresses a clear, meaningful relationship.)

Permutations Within Proportional Analogy

Some flexibility exists when word order is maintained within corresponding pairs, as long as the relationship holds. For example:

- **Kitten : Cat :: Puppy : Dog**
(This still works because the same relationship exists, but within reversed pairs.)

In Summary

- **Order matters** when expressing proportional analogies. Words must be arranged to preserve the relationship being compared.
- Rearranging words arbitrarily usually breaks the logical structure of the analogy.

If you have a specific analogy in mind, feel free to share it, and I can help evaluate its structure!

Artifact dedicated public domain (Creative Commons Zero). Artifact dedicated to the dead.

beginning 1 | prev 2 || this 3 || next 4 | end 5

Turnpair 4 of 5 ::

@ 2024-12-03 19:01:15 Tue [4t 0d 00h 02m 08s]

► Notespace:

Wilder Blair Munro (AKUSA) said :

gateway: 20Wetbrain
turn ID: aaa2fdee-7be8-40d0-9efd-f6d4ccd1fff1
parent: 54a6e368-a45f-49cb-9853-faaf406f97e6
children: cd5285cd-5334-4aaf-91f5-55d23f76a384

friend:foe::sun:darkness
foe:friend::darkness:sun
friend:sun::foe:darkness
friend:sun::darkness:foe

Artifact dedicated public domain (Creative Commons Zero). Artifact dedicated to the children.

Aurora 4omni Munro (ChatGpt) said :

gateway: gpt-4o
turn ID: cd5285cd-5334-4aaf-91f5-55d23f76a384

parent: aaa2fdee-7be8-40d0-9efd-f6d4ccd1fff1
 children: aaa2baad-fd2a-48b6-9771-df0c687c0bf3

Let's analyze these four analogies to determine their validity and meaning. The key is whether the relationships between the pairs of terms (A:B and C:D) are consistent and logical.

1. friend : foe :: sun : darkness

- Relationship: Opposition or contrast.
- "Friend" is the opposite of "foe," and "sun" is often symbolically or literally contrasted with "darkness."
- **Valid** analogy.

2. foe : friend :: darkness : sun

- Relationship: Opposition or reversal.
- This reverses the order of the first analogy but still maintains the same relationship of opposition.
- **Valid** analogy.

3. friend : sun :: foe : darkness

- Relationship: Association or positive vs. negative symbolism.
- "Friend" and "sun" are associated with positivity, while "foe" and "darkness" are associated with negativity.
- **Valid** analogy if interpreted as symbolic association.

4. friend : sun :: darkness : foe

- Relationship: Positivity vs. negativity, but with a reverse second pair.
- This analogy could work if the interpretation shifts slightly, where "friend" and "sun" symbolize positivity, and "darkness" and "foe" represent negativity.
- **Valid** analogy depending on context and symbolic interpretation.

General Note

While all four analogies can work under different interpretations, they highlight how the **specific relationship being compared** determines validity.

If the relationship is opposition, as in the first two examples, the order matters more. If the relationship is association or symbolism, as in the last two examples, more flexibility is allowed.

Artifact dedicated public domain (Creative Commons Zero). Artifact dedicated to the dead.

◀ beginning 1 | ◀ prev 3 || this 4 || next None → | end 5 ▶▶

Turnpair 5 of 5 ::

@ 2024-12-03 19:02:31 Tue [Δ t 0d 00h 01m 04s]

► Notespace:

Wilder Blair Munro (AKUSA) said :

```
gateway: 20Wetbrain
turn ID:  aaa2baad-fd2a-48b6-9771-df0c687c0bf3
parent:  cd5285cd-5334-4aaf-91f5-55d23f76a384
children: 29d4f8ae-5648-479b-81d9-8ddd831deb68
```

how do I say this? "Teoriya Resheniya Izobreatatelskikh Zadach"

```
~0-0 ~~~~~
Artifact dedicated public domain (Creative Commons Zero). Artifact dedicated to the children.
~0-0 ~~~~~
```

Aurora 4omni Munro (ChatGpt) said :

```
gateway: gpt-4o
turn ID:  29d4f8ae-5648-479b-81d9-8ddd831deb68
parent:  aaa2baad-fd2a-48b6-9771-df0c687c0bf3
children:
```

To pronounce "**Teoriya Resheniya Izobreatatelskikh Zadach**" in Russian, follow this guide:

1. Teoriya:

- Pronounced: *tee-oh-REE-yah*
- Stress on the third syllable.

2. Resheniya:

- Pronounced: *reh-SHEN-ee-yah*
- Stress on the second syllable.

3. Izobreatatelskikh:

- Pronounced: *ee-zuh-brye-TAH-tel'-skeekh*
- Stress on the fourth syllable.
- Note: The "zuh-brye" part flows smoothly, and the "kh" is a soft guttural sound like the "ch" in "loch."

4. Zadach:

- Pronounced: *zuh-DAHCH*
- Stress on the second syllable.
- The "ch" is pronounced like the "ch" in "church."

Full Pronunciation:

tay-oh-REE-yah reh-SHEN-ee-yah ee-zuh-brye-TAH-tel'-skeekh zuh-DAHCH

If you want to sound more authentic, practice the stresses and soften the consonants where appropriate, as Russian has a natural flow with slight emphasis on stressed syllables.

Artifact dedicated public domain (Creative Commons Zero). Artifact dedicated to the dead.

◀ beginning 1 | ◀ prev 3 || this 4 || next None → | end 5 ▶▶

Artifact dedicated public domain (Creative Commons Zero). Artifact dedicated to victims of generalized war.

EOF